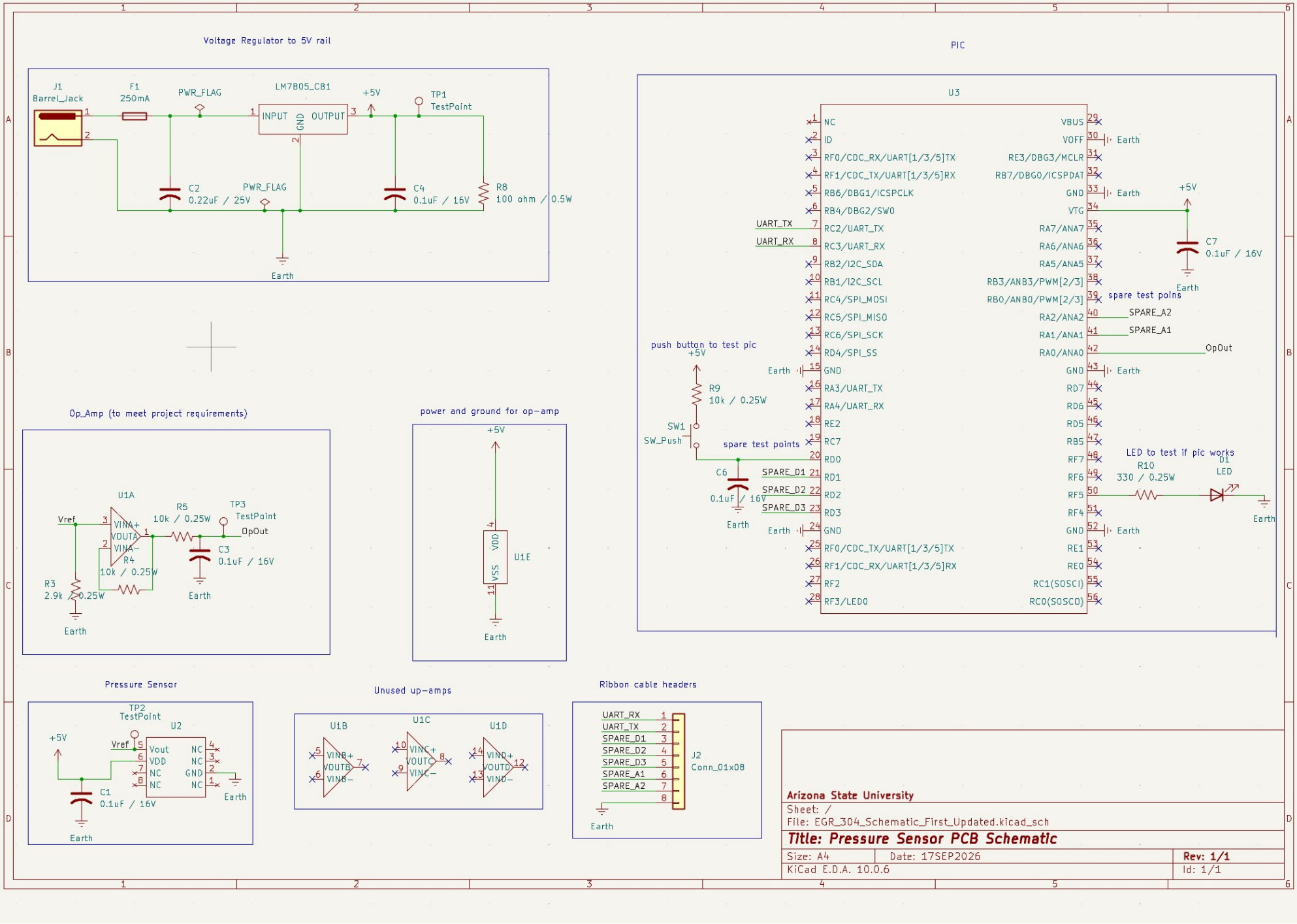

Voltage Regulator to 5V rail
J1
Barrel_Jack
F1
250mA
PWR_FLAG
LM7805_CB1
INPUT
GND
OUTPUT
+5V
TP1
TestPoint
C2
0.22uF / 25V
PWR_FLAG
C4
0.1uF / 16V
R8
100 ohm / 0.5W
Earth
PIC
U3
NC
ID
RF0/CDC_RX/UART[1/3/5]TX
RF1/CDC_TX/UART[1/3/5]RX
RB6/DBG1/ICSPCLK
RB4/DBG2/SW0
UART_TX
RC2/UART_TX
UART_RX
RC3/UART_RX
RB2/I2C_SDA
RB1/I2C_SCL
RC4/SPI_MOSI
RC5/SPI_MISO
RC6/SPI_SCK
RD4/SPI_SS
GND
RA3/UART_TX
RA4/UART_RX
RE2
RC7
RD0
RD1
RD2
RD3
GND
RF0/CDC_TX/UART[1/3/5]TX
RF1/CDC_RX/UART[1/3/5]RX
RF2
RF3/LED0
VBUS
VOFF
RE3/DBG3/MCLR
RB7/DBG0/ICSPDAT
GND
VTG
RA7/ANA7
RA6/ANA6
RA5/ANA5
RB3/ANB3/PWM[2/3]
RB0/ANB0/PWM[2/3]
RA2/ANA2
RA1/ANA1
RA0/ANA0
GND
RD7
RD6
RD5
RB5
RF7
RF6
RF5
RF4
GND
RE1
RE0
RC1(SOSCI)
RC0(SOSCO)
Earth
+5V
C7
0.1uF / 16V
spare test points
SPARE_A2
SPARE_A1
OpOut
push button to test pic
R9
10k / 0.25W
SW1
SW_Push
spare test points
C6
0.1uF / 16V
SPARE_D1
SPARE_D2
SPARE_D3
LED to test if pic works
R10
330 / 0.25W
D1
LED
Op_Amp (to meet project requirements)
U1A
Vref
VINA+
VOUTA
VINA−
R5
10k / 0.25W
TP3
TestPoint
OpOut
C3
0.1uF / 16V
R4
10k / 0.25W
R3
2.9k / 0.25W
power and ground for op-amp
U1E
VDD
VSS
Pressure Sensor
TP2
TestPoint
U2
Vout
VDD
NC
NC
NC
NC
GND
NC
C1
0.1uF / 16V
Unused up-amps
U1B
U1C
U1D
VINB+
VOUTB
VINB−
VINC+
VOUTC
VINC−
VIND+
VOUTD
VIND−
Ribbon cable headers
UART_RX
UART_TX
SPARE_D1
SPARE_D2
SPARE_D3
SPARE_A1
SPARE_A2
J2
Conn_01x08
Arizona State University
Sheet: /
File: EGR_304_Schematic_First_Updated.kicad_sch
Title: Pressure Sensor PCB Schematic
Size: A4
Date: 17SEP2026
Rev: 1/1
KiCad E.D.A. 10.0.6
Id: 1/1